Sheet 57 Thu. 1 st Apr. 2027 Determinant and equations of lines: review

1. Warm-up · Flash Slope of the line 3 $x + 2y - 6 = 0$?
2. Core · Method Equation of the line (AB) with $A(1;2)$ and $B(4;8)$.
3. Core + · Method Are the lines 2 $x - y + 1 = 0$ and $4x - 2y + 5 = 0$ parallel, coincident or intersecting?
4. Back · Flash Calculate $10^{-2} \times 5 \times 10^{4}$.
5. Running thread · Method Solve in $\mathbb{R}$: $7 - 2(x - 1) = 3x - 6$.

Sheet 58 Mon. 5 Apr. 2027 Plane geometry problems: choosing a method

1. Warm-up · Flash Distance AB for $A(0;0)$ and $B(5;12)$?
2. Core · Method $A(0;0)$, $B(4;2)$, $C(-2;4)$. Type of triangle ABC ?
3. Core + · Method Give two methods to prove that $ABCD$ is a parallelogram using coordinates.
4. Back · Reading In a class of 30 pupils, a bar chart shows x pupils with mark 12 and $2x$ pupils with mark 14; the other marks account for 18 pupils. Find x.
5. Running thread · Flash A pyramid has a square base of side 6 cm and height 5 cm. Calculate its volume.

Sheet 59 Tue. 6 Apr. 2027 Situations from other disciplines

1. Warm-up · Flash A moving point has position $x(t) = 3t + 2$ (in m, t in s). Where is it at $t = 4$?
2. Core · Method A second moving point follows $x_2(t) = 5t - 4$. At what time and where do the moving points meet?
3. Core + · Flash A velocity vector has coordinates $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ (in m/s). What is its magnitude?
4. Back · Flash Calculate $\frac{1}{2} - \frac{3}{5}$.
5. Running thread · Write up ABC is isosceles at A and H is the foot of the altitude from A. What can be stated of H ?

Sheet 60 Thu. 8 Apr. 2027 Communicating a proof: implication, converse, equivalence

1. Warm-up · Flash State the converse of: "if a quadrilateral is a rhombus, then its diagonals are perpendicular".
2. Core · True/false True or false? Justify. The previous converse is true.
3. Core + · Write up The equivalence " $x^2 = 4 \Leftrightarrow x = 2$" correct? Correct it.
4. Back · Flash Compare 2^{10} and 10^3.
5. Running thread · Flash Between which consecutive integers does $\frac{7}{4}$ lie on a number line? Give its decimal value.

Sheet 61 Mon. 26 Apr. 2027 Oral class discussion: comparing methods

1. Warm-up · Flash Distance between $A(-2;1)$ and $B(1;5)$?
2. Core · Method Give three methods to prove that A, B, C are collinear.
3. Core + · Reasoning To prove that two lines given by $y = m_1x + p_1$ and $y = m_2x + p_2$ are parallel, which method is quickest?
4. Back · Factorise Factorise $6x + 9$.
5. Running thread · Method Write 360 as a product of primes. How many divisors does it have?

Synthesis problems, reasoning and automatic skills